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RESEARCH ARTICLE

Adaptive and Interpretable Speed Control Rules for En Route Arrival Management Using Extended Learning Classifier System

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ABSTRACT Air-transport infrastructure is facing increasing congestion, necessitating innovative approaches to optimize arrival management and minimize delays near airports. To address this issue, this study developed an en route arrival management framework that employs a learning classifier system (LCS) to reduce arrival delays and fuel consumption by generating adaptive and interpretable speed-control rules. A virtual design database, created from multiobjective optimization and integrated with a cellular automaton (CA)-based air-traffic model, facilitates the development of these rules. Within this database, a binary target variable represents speed-control decisions, whereas explanatory variables encompass factors such as aircraft proximity and congestion levels. The LCS is trained on this dataset to generate speed-control rules, and their effectiveness is evaluated through CA-based traffic simulations. The results indicate that the LCS achieved 96.8% accuracy in predicting appropriate speed-control actions, significantly outperforming traditional decision-tree methods (70.5% accuracy). Furthermore, experimental findings demonstrate reductions in average flight times by 10–20 s and up to 5 fewer spacing adjustments for traffic within 600 s before and after the arrival of speed-controlled flights. Overall, the LCS reduced total flight time by 1,801 s and total fuel consumption by 1,650.1 kg per operational cycle, resulting in an estimated annual cost savings of $466,000. Furthermore, the LCS approach successfully extracts both sector-specific and common rules, offering enhanced adaptability to various traffic scenarios. These results suggest that the ability to generalize rules across different airspace sectors can improve the safety and efficiency of air-traffic management.

INDEX TERMS Air traffic management, arrival management, machine learning, learning classifier system.

I. INTRODUCTION

A. MOTIVATION

Air transport has been a catalyst for economic growth and social development through connectivity, trade, and tourism. Following a downturn due to COVID-19, air traffic has rebounded, with 4.5 billion passengers and USD 908 billion in revenue generated in 2023 [1]. Despite this growth, cargo revenues have decreased and operating costs have increased, primarily owing to higher fuel prices. The aviation industry supports 87.7 million jobs globally [2] and passenger traffic is expected to double by 2040 [3]. To accommodate this increasing demand, various countries have launched air-traffic management (ATM) initiatives, such as CARATS by Japan [4], NextGen by the USA [5], and the SESAR by the EU [6]. However, reliance on outdated technology causes delays. For instance, in 2019, 19.9% of US and 23.5% of European flights were delayed by more than 15 min on average [7]. This emphasizes the importance of integrating flow management approaches into arrival management to

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ensure a balance between air-traffic demand and available capacity across both en route airspaces and airport operations.

Arrival Manager (AMAN) [8] is a system developed to aid air-traffic controllers (ATCos) in optimizing the flow of arriving flights, thereby minimizing the need for airborne holding and improving runway throughput. AMAN has been deployed at major airports worldwide since its first implementation in Zürich in 2001. Strategies for en route arrival management, such as Extended AMAN (E-AMAN) and En Route AMAN, have been developed to enhance arrival management [9] by complementing the existing AMAN. E-AMAN coordinates aircraft sequencing and spacing from the terminal maneuvering area (TMA) to the en route airspace, whereas En Route AMAN handles speed adjustments and runway assignments to alleviate congestion before aircraft reach the TMA. Previous studies explored en route speed control before the top of the descent to mitigate TMA congestion, focusing on fuel savings [10] and delayed absorption [11]. Other approaches have integrated speed control with ground and airborne delay management, allowing for flexible delay management [12].

Although en-route arrival management is being gradually implemented, predicting flight times remains challenging owing to the complexity of models, data usage, and the need for precise inputs, such as aircraft performance and weather conditions. Although ensemble learning has improved accuracy, most support systems focus on terminal airspaces, making flight time prediction in en route airspaces particularly challenging [13]. Furthermore, managing arrival sequences in en route airspaces is complicated by the complex network of intersecting airways and traffic flows, increasing congestion and demanding real-time traffic adjustments that increase the ATCo workload and operational risks [14]. While traditional AMAN systems, including E-AMAN, aim to optimize arrival sequences through real-time adjustments, they often struggle to accommodate the uncertainties inherent in dynamic traffic conditions. This limitation necessitates substantial intervention from ATCos to maintain safety and efficiency. Furthermore, the current E-AMAN system, which extends traditional time-based methods, may not align well with the distance-based deconflicting operations commonly employed in en-route airspaces, potentially reducing their acceptance among controllers. Therefore, an en route arrival management system that can be easily integrated with current practices is crucial to ensuring successful implementation. This study addresses these challenges by introducing an adaptive en route arrival management framework using an extended learning classifier system (XCSI). Our approach derives interpretable and adaptable speed control rules offline, eliminating the need for real-time computations and facilitating more scalable and flexible solutions for dynamic air-traffic management.

B. STATE-OF-THE-ART SYSTEMS

1) E-AMAN AND EN ROUTE AMAN

Various studies have explored delay absorption strategies for E-AMAN and En Route AMAN, including speed control, rerouting, and runway scheduling, to distribute delays upstream and minimize disruptions in the TMA.

Khassiba et al. [15] introduced a two-stage stochastic programming model for aircraft arrival management that incorporates multiple initial approach fixes and considers various flight statuses, thereby enhancing the precision of cost evaluations in scheduling. Jun et al. [16] proposed dynamic speed control strategies to reduce aircraft delays by integrating holding detection with machine-learning techniques. They examined the effects of varying levels of collaboration on these strategies, focusing on Singapore's multi-Air Navigation Service Provider system. Huo et al. [17] proposed a dynamic E-AMAN that incorporates speed control, route changes, runway assignments, and TMA-entry speed adjustments. Their model updates instructions from 500 NM out, using real-time safety weights and simulated annealing with a rolling horizon for continuous optimization. Sáez et al. [18] incorporated a continuous descending operation and introduced a method for managing arrival traffic in the TMA using two areas: a pre-sequencing area for 4D trajectory negotiation and a dynamic-trajectories area for generating adaptable arrival routes. Similar to [18], Kamo et al. [19] proposed a holistic framework that combines trajectory optimization and sequencing into a single process. They incorporated weather forecast uncertainties using ensemble forecasts to enhance decision-making robustness. For weather forecasts, Temme et al. [20] introduced an AMAN that integrates real-time severe weather data to improve ATM. This approach improves situational awareness and decision-making during adverse weather using advanced visualization and predictive analytics.

Although these state-of-the-art AMAN systems using optimization techniques effectively transfer delays from the TMA to the upstream airspace, several common challenges still persist in aircraft arrival management. One primary concern is the management of uncertainty and conflict. As highlighted by Huo et al. [17] and Kamo et al. [19], uncertainties in the estimated time of arrivals (ETAs) used to stipulate the scheduled time of arrivals owing to weather forecasts and flight disturbances significantly impact operational efficiency. These uncertainties can alter the delay duration and sequence during congestion, which in turn increases the workload of ATCos. For instance, unpredictable delays or changes in aircraft arrival times require ATCos to constantly adjust and optimize their strategies, resulting in a higher number of radar and tactical planning tasks. This increased workload affects the effectiveness of implementing En Route AMAN. To address these uncertainties, a recent study optimized arrival scheduling by formulating the traveling salesman problem under uncertainty by introducing gradient-boosting machine learning and quantile regression for ETA prediction [21].

Other studies have proposed stochastic approaches using queuing theory models [22], [23], [24], [25], [26], [27], which help identify airspace bottlenecks [23], [24], [25] and develop conducive interarrival control [26], [27] for En Route AMAN. The $M/G/1$ queuing model demonstrated that congestion pricing can improve airport operational efficiency by timing aircraft arrivals and encouraging them to arrive at off-peak times, potentially enhancing overall service levels [22]. With the most efficient transition from flow management to time-based operations occurring at a 70 NM point from the airport [23], the $G/G/c$ queuing model demonstrates that arrival management, which effectively reduces the time-interval variance between preceding aircraft, significantly reduces the total arrival delay [24]. The $M/G/c/K$ queuing model [25] revealed that reducing the variance in arrival intervals to within 100 s, which corresponds to speed control in an en route airspace more than 150 NM away from the arrival airport, is more effective in reducing delays than increasing the TMA capacity by 10%. Compared to these time-invariant models, the $G_t/GI/s_t + GI$ queuing model demonstrated, both theoretically [26] and practically through fast-time simulations [27], that TMA congestion can be largely mitigated by smoothing the arrival rate at the terminal entry fix using nonlinear integer programming, while adhering to operational constraints. Building on this concept of interarrival control, algorithms incorporating runway assignments and sector-specific speed control, based on data-driven and data exploration methods, have been proposed [28] and validated through human-in-the-loop simulations [29].

Although a previous study [28] successfully implemented speed control based on rule sets derived through decision-tree analysis, this approach exhibited limitations in capturing the full complexity of dynamic air-traffic scenarios. To address these limitations, this study extends the previous approach by employing alternative data exploration techniques to enhance the speed control algorithm. This new approach aims to generate more flexible and adaptive rules that better accommodate the variability inherent in air-traffic conditions, ultimately enhancing the overall performance of speed control strategies.

2) DATA EXPLORATION

Data exploration is a crucial initial step in data analysis, involving the analysis of datasets to uncover underlying patterns, trends, and relationships. It employs techniques such as visualization, statistical analysis, and querying to comprehensively understand the structure and characteristics of data. The primary goals are to extract meaningful insights that can guide decision-making, identify key variables, and support further analysis, such as model building or rule development. By revealing the dynamics inherent in the data and refining hypotheses, data exploration ensures that subsequent analysis or predictive modeling is grounded in relevant and meaningful information.

Several methods have been employed to quantitatively and objectively extract design information from the objective and design variables of optimal solutions. These include genetic programming (GP), rough set theory, association rules, and decision-tree analysis. Bandaru et al. [30] proposed a GP-based method to automatically extract important mathematical relations from Pareto-optimal solution sets. Greco et al. [31] proposed the dominance-based rough set approach that guides a preference model expressed in terms of IF-THEN-based decision rules using a sample of solutions from a Pareto-optimal set. Association rules are similar to rough set theory but offer the advantage of extracting rules based on confidence. Sugimura et al. [32] proposed a data exploration approach that combines robust multiobjective optimization and an association rule with an aspiration vector to analyze the trade-off relationships. Dudas et al. [33] generated different rule sets for different objectives of a real-world flexible production line optimization problem. However, a common drawback of these approaches is that, although they allow for visual interpretation of the learning model, the artificial manipulation required for such interpretation may result in a loss of both quantitativity and objectivity.

Learning Classifier Systems (LCSs) [34] are a class of machine-learning methods that use interpretable learning models. This interpretability stems from the fact that decision-makers can directly understand the underlying learning model, which retains quantitative information without the need for artificial manipulation. Essentially, the interpretable learning model itself becomes a source of quantitative knowledge, contributing to objective understanding. Evolutionary machine learning, a technique that integrates evolutionary computation into machine learning, plays a crucial role in LCSs. In this process, evolution generates interpretable learning models that promote adaptation to the environment, whereas the learning component evaluates these models. LCS-based data mining has been employed in previous studies. Kharbat et al. [35] applied an accuracy-based LCS (XCS) [36] to analyze the Frenchay Breast Cancer dataset, grouping the resulting rules using the quality-threshold clustering algorithm [37] to provide insightful feedback to medical experts. Their results demonstrated that the XCS outperformed C4.5 [38], a well-known decision-tree induction learning technique, in knowledge discovery. Iqval et al. [39] leveraged LCS for object recognition by developing an extraction technique for image feature points. They showcased the capacity of LCSs to autonomously learn different feature combinations for various image types, with evolving generalized rules for computing saliency maps outperforming individual feature-based and seven combination methods in detecting salient objects from three different types of benchmark datasets with varying difficulties. Chiba and Nakata [40] proposed an LCS to obtain optimal solutions for multiobjective optimization problems using the design of hybrid rocket engines as a case study. They reported that LCS can generate rules that account for the degree of improvement

in the multiobjective function or, more precisely, the accuracy of the objective function value of the dataset. Arif et al. [41] applied LCSs to extract spam-detection rules from text, addressing the sparsity of feature vectors generated using the term frequency, inverse document frequency, and the sentiment vocabulary of word *n*-grams. They introduced a new encoding scheme to represent the classifier rules, smoothening the learning process and producing favorable results.

As discussed above, the real-world applications of LCSs for data mining are limited. Additionally, to the best of our knowledge, there are no examples of their application in ATM. Given their ability to extract superior knowledge and exhibit superior learning performance compared to other data-mining methods, LCSs hold significant promise for generating interpretable speed control rules that are highly generalizable to diverse traffic situations. This potential for generating adaptable and transparent rules motivates our investigation into the application of LCSs for optimizing en-route arrival management.

C. STUDY CONTRIBUTIONS AND PAPER STRUCTURE

This paper presents a novel LCS-based approach for arrival management, with the aim of minimizing delays near destination airports through adaptive speed control and rule-based decision-making. The main contribution is the **development of a framework for analyzing speed control rules in En Route AMAN based on relative spacing and traffic flow using the XCSI** [42]. This framework includes a detailed algorithmic design, validation of the learning performance, and in-depth analysis of the extracted rules. The overall objective is to distill complex speed control rules into simple, interpretable rules that can inform and support decision-making in ATM.

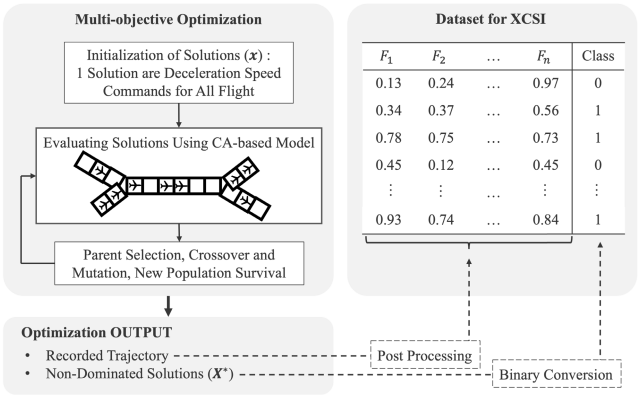


FIGURE 1. Dataset construction framework.

The remainder of this paper is structured as follows: Section II describes the virtual design dataset used in our experiments, focusing on the key relevant features to ATM operations. Section III presents the XCSI and explains its application to en route arrival management, with the aim of extracting interpretable speed control. Using the introduced dataset and method, Section IV details the training of the speed control decision model using XCSI. Section V describes its implementation in a CA-based simulator [43] and evaluates its performance by comparing it with those obtained in previous studies [28], highlighting the improvements in rule generation and performance metrics. Section VI further analyzes the operational impacts of the extracted rules and examines their applicability in a trained traffic scenario (Section VI). Finally, VII discusses the key findings and their implications for ATM practices, and Section VIII offers concluding remarks.

II. DATASET DESCRIPTION

A. DATASET GENERATION THROUGH MULTIOBJECTIVE OPTIMIZATION AND CELLULAR AUTOMATON(CA)-BASED AIR TRAFFIC MODEL

Fig. 1 illustrates the dataset construction framework, which employs multiobjective optimization and a CA-based air traffic model [28], [43]. In this approach, the LCS generates adaptive rules during the learning phase by aligning them with the non-dominated solutions obtained through multiobjective optimization. The CA-based air traffic model serves as an evaluator within the optimization framework. After obtaining the non-dominated solutions, a post-processing step is performed using local traffic information derived from the recorded trajectories. Specifically, the decision of whether to apply speed control is determined based on the non-dominated solutions, and a binary variable is assigned to each aircraft, indicating whether speed control was executed. These variables serve as class labels, whereas the corresponding traffic features extracted from the recorded trajectories are used as input attributes for the XCSI dataset.

B. ARRIVALS AT TOKYO INTERNATIONAL AIRPORT (RJTT)

Fig. 2 illustrates the radar tracks of the inbound flights to RJTT under northerly wind conditions and following standard traffic patterns. Four distinct clusters are shown: Clusters 1–3 account for 472.7 daily flights arriving from the southwest, predominantly landing on runway 34L, whereas Cluster 4 (depicted in purple) comprises 151.2 daily flights from the north, landing on runway 34R. Thus, runway 34L accommodates more than three times as many arrivals as 34R. Cluster 3 (light blue) includes three daily domestic flights from RJTH and five international flights from the Pacific Ocean to the east and Oceania to the south. Clusters 1 and 2, which account for the majority of southbound arrivals (306.1 and 158.3 flights respectively), were the focus of our analysis, specifically the cruise speed control within sectors T25 and T24 (highlighted in red in Fig. 2) and T14 (shown in green), 150–200 NM from RJTT. These clusters also experience disruptions from 59.7 daily "pop-up" flights—departures from airports located within 250 NM of RJTT—that merge with the flow of incoming traffic. Notably, pop-up flights from RJOO (29.9 flights/day) and RJBB (13.2 flights/day) necessitate spacing adjustments as

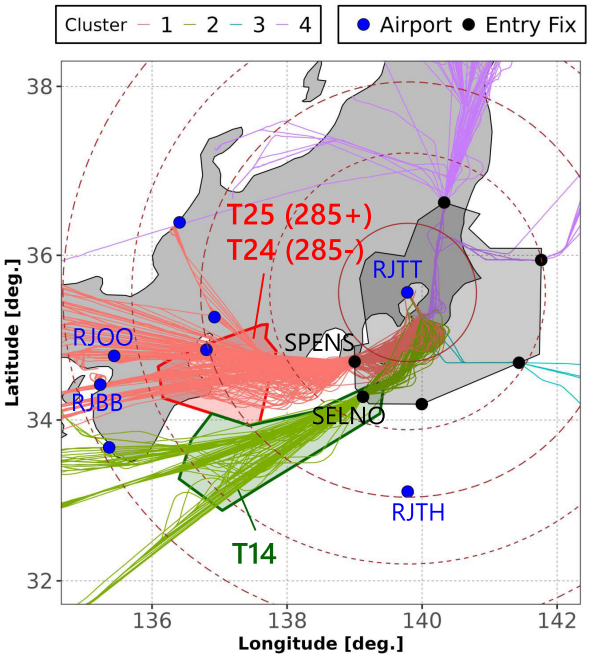


FIGURE 2. Actual radar tracks color-coded according to the direction of approach to the RJTT during northerly wind operations on a day in December 2019 [28]. The Tokyo Approach Control Area is outlined by a black polygon, with six entry points marked by black dots. The brown dotted circle with a 50 NM radius is centered on RJTT. The red and green polygons represent en route sectors where cruise speed control and runway assignment decisions are made. Airports within 250 NM of RJTT are indicated by blue dots. RJTT, Osaka International Airport (RJOO), Kansai International Airport (RJBB), and Hachijojima Airport (RJTH) are labeled.

they merge with the Cluster 1 traffic (depicted in orange in Fig. 2).

Another key feature of Clusters 1 and 2 is the path extension before entering RJTT terminal airspace or Tokyo Approach Control Area (TACA) to mitigate overdemand. The entry points for these clusters are SPENS and SELNO (marked with black dots in Fig. 2).

C. MINIMIZING FLIGHT TIME USING MULTIOBJECTIVE OPTIMIZATION

To generate data for the XCSI, a multiobjective optimization approach, combining the non-dominated sorting genetic algorithm II (NSGA-II) [44] with a cellular automata (CA)-based air-traffic model [43], was employed. This approach attempts to derive suboptimal solutions for cruise speed control that alleviate congestion in the TMA without significantly increasing the overall flight time, which could negatively impact punctuality. The impact of "pop-up" flights, particularly those merging from the south, were also considered owing to their significant disruptive influence on cruise traffic [45].

The optimization process employs two objective functions to minimize the average flight times for both (1) pop-up flights (f_1) and (2) cruise inflow flights (f_2) from the south. The optimization parameters and settings adhere to the methodology detailed in [43], ensuring sufficient convergence through a larger population size and higher number of evaluations. The deceleration values are based on aerodynamic and structural constraints, with 435 design variables x_i representing the deceleration applied to each aircraft in the cruising state. Further details regarding the optimization process can be found in [28].

D. TARGET AND EXPLANATORY VARIABLE

This study uses Pearson's correlation coefficient r [46] to analyze the strategies that minimize f_2. Non-dominated solutions similar to the optimal solution $\boldsymbol{x}^* \in \mathbb{R}^N$, including each flight i's speed command x_i^*, are selected for further analysis. The design variables x_i are converted into binary values, where 0 and 1 indicate cruising speed and speed control for use as class labels, respectively.

TABLE 1. Description of feature F.

Feature	Description
FlightCategory	0: pop-up flights, 1: cruise flights.
Time	Time of speed-control decision
Alt	Altitude [ft]
GS	Ground speed [kt]
TAS	True airspeed [kt]
Wind	Wind speed in the traveling direction [kt]
RouteNum	Route ID assigned at the speed-control points
DtAC_i	Distance between the aircraft and i-th aircraft ($i \leq 10$) ahead of it [NM] (see Fig. 3)
NumAC	# of aircraft ahead of the aircraft (see Fig. 3)
IsCluster1	0:T14 traffic, 1:T25 traffic (see Fig. 2).

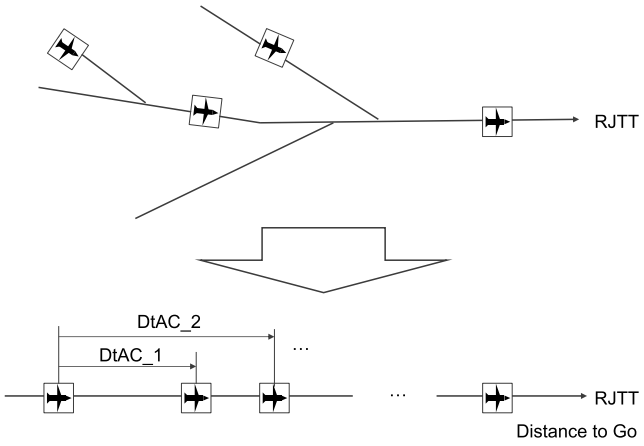


FIGURE 3. Description of DtAC_i, which is calculated by considering all the routes including those in Clusters 1 and 2.

Table 1 and Figure 3 illustrate the 19 selected features, such as flight category, altitude, ground speed, and distance between aircraft. The dataset includes key features, such as ground speed, altitude, and distance to the preceding aircraft (DtAC), that are crucial for determining speed-control actions. For example, maintaining a safe separation distance (DtAC) is essential for reducing collision risks, whereas ground-speed adjustments help manage approach sequencing

and minimize congestion near the TMA. Additionally, wind speed data were included to account for the environmental factors affecting aircraft performance. This study introduces a novel feature, " *IsCluster1*," which leverages XCSI's generalization capability through the use of "#." This enabled the extraction of both shared and cluster-specific rules based on different flight directions. For further details on the optimization process and analysis, refer to [28].

Having established an understanding of the dataset and its key features, we now describe the data-mining methodology employed to extract interpretable rules for speed control using an LCS and its extension XCSI.

III. DATA-MINING METHODOLOGY

A. LCS

This study employed the LCS [34], a genetic machine-learning technique that incorporates evolutionary computation. LCSs generate IF–THEN rules, called classifiers, which are optimized for effectiveness within their environment and learning through continuous evaluation and refinement. Specifically, the LCS performs actions based on the current state of the environment and learns based on the resulting rewards. Furthermore, the LCS uses a genetic algorithm to create and delete classifiers, evolving a set of rules that are increasingly well-suited for the environment.

A unique feature of LCSs is their inherent ability to generalize. As it is impractical to experience and learn behaviors in every possible state, LCSs generalize from experience, assuming that the same behavior is experienced in other states. This capability addresses the "curse of dimensionality," wherein learning becomes more difficult as the number of environmental states increases. Specifically, LCSs achieve generalization by incorporating wildcards (#) in the conditional part of the rules, allowing us to obtain a generic classifier that matches multiple states (inputs).

In this study, the generated IF–THEN rules, derived flight data, provide ATCos with a transparent understanding of the logic behind speed control decisions. Unlike traditional black-box models, LCSs produce readily interpretable rules that can be directly applied in real-time operations. The generalization ability of the algorithm ensures that these rules can adapt to diverse traffic conditions encountered in en route airspaces.

Although various advancements have been made in LCS technology, this study uses XCSI [42], which is adaptable and can handle real numbers. Before explaining XCSI, we first describe the basic mechanisms of the XCS [36], where conditional statements comprise binary values, before explaining the extension to real-valued conditions in XSCI.

B. ACCURACY-BASED LEARNING CLASSIFIER SYSTEM (XCS)

As shown in Fig. 4, XCS is a type of LCS that learns to maximize the sum of REWARD obtained from the environment by performing the appropriate action ACTION on a STATE, which is input from the environment. The DETECTOR converts the state of the environment into an internal representation of XCS, whereas the EFFECTOR translates its internal actions back into environmental actions.

The following sections delve deeper into the XCS mechanism, covering (1) representation, (2) performance component, (3) update, (4) discovery component, and (5) real-value extension.

1) REPRESENTATION

As shown in Fig. 4, the classifiers (cl) within the XCS population ($[P]$) are represented by wildcards (#), which denote discrete or arbitrary values. For example, 11#0 represents a conditional part with four conditions, where "#" can match any value. The action components can be expressed arbitrarily. To combine classifiers that have the same condition and action part into a single classifier, the *Numerosity* parameter is retained. Each classifier also includes the *Prediction*, *Error*, and *Fitness* parameters. In the following, the i-th classifier is denoted as cl_i, with its corresponding *Numerosity*, *Prediction*, *Error*, and *Fitness* parameters denoted as num_i, p_i, ϵ_i, and F_i, respectively.

The population is initialized with a maximum number of classifiers (N), comprising N randomly generated classifiers for the condition and action parts. The initial values of p_i, ϵ_i, and F_i are denoted as p_I, e_I, and F_I.

2) PERFORMANCE COMPONENT

The performance component selects and sends an appropriate action to a state that refers to a set of classifiers. When state x is the input from the environment, the condition part matches the state against the classifiers represented by $[P]$ in Fig. 4. During the matching, it is checked whether the classifiers match exactly, except for # in the condition part (C). The matching classifiers form a match set represented by $[M]$ in Fig. 4, as follows:

$$[M] = \{cl \in [P] | x \in cl.C\} \tag{1}$$

If no matching classifier exists in the matching set, a covering mechanism is executed to create a new classifier that matches the input state. Next, for each possible action $a_j \in A$, a subset $[M]|a_j$ of $[M]$ is generated as follows:

$$[M]|a_j = \{cl \in [M] | cl.A = a_j\} \tag{2}$$

In this covering mechanism, a condition part is created to match the input state, but with a certain probability ($P_\#$) of including # in each condition. The action part of the classifier in the collation group may contain multiple competing actions. To resolve this issue, a prediction array is created by linking the predicted gains from the competing behaviors. The predicted reward $P(s, a_j)$ for each action a_j in the prediction array is calculated as follows:

$$P(x, a_j) = \sum_{cl_k \in [M]|a_j} p_k \times \frac{F_k}{\sum_{cl_k \in [M]|a_j} F_k} \tag{3}$$

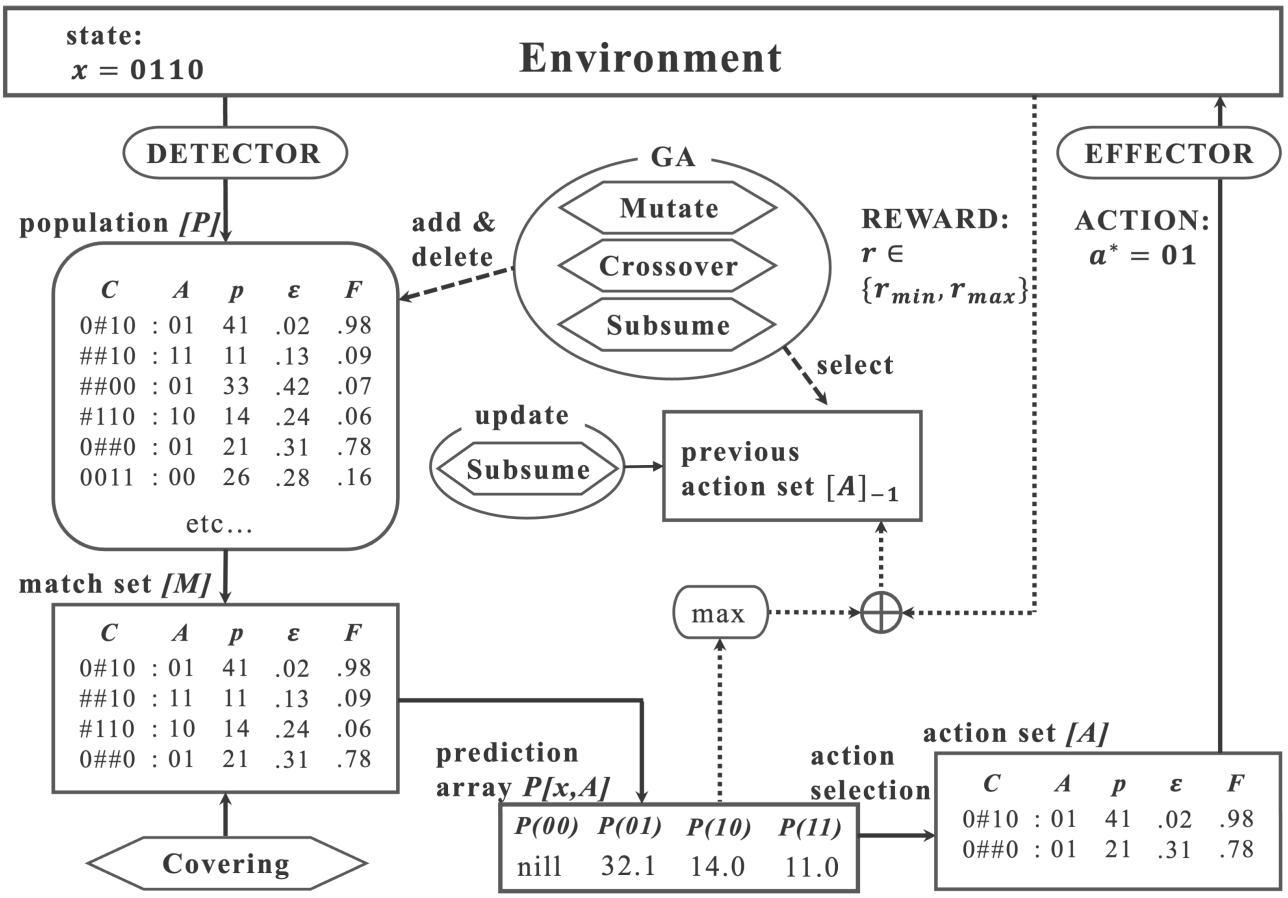


FIGURE 4. XCS framework.

where the XCS selects and executes action a^* and obtains reward $r \in \{r_{min}, r_{max}\}$ based on a certain strategy (random selection or ϵ-greedy selection). The classifiers with selected action a^* in the action part form the action set represented by $[A]$ in Fig. 4. The series of processes in the execution part is completed once the selected action is executed. This processing sequence is defined as one step.

3) UPDATE

After the performance component in each step is completed, the update part renews the predictions, error values, and fitness of the classifiers in the action set based on the reward obtained from the environment. This process is performed for the previous action set represented by $[A]_{-1}$ in Fig. 4. The update involves applying the following renewal rule to each classifier cl_i (the left arrow ($\leftarrow$) indicates assignment):

$$P \leftarrow r + \gamma max_a P(x, a) \tag{4}$$

$$p_i \leftarrow p_i + \beta(P - p_j) \tag{5}$$

$$\epsilon_i \leftarrow \epsilon_i + \beta(|P - p_j| - \epsilon_j) \tag{6}$$

where P is the target value for updating the prediction p_i calculated using the reward r from the previous step and the maximum value $max_a P(x, a)$ from the prediction array. Additionally, β and γ are the same as the learning and discount rates in reinforcement learning methods, respectively, indicating the speed of performing fitness and amount of future reward considered, respectively.

Thereafter, based on the updated error value ϵ_i, the absolute accuracy κ_i of the classifier and the relative accuracy κ_i' in $[A]_{-1}$ are calculated, and the fitness F_i is obtained as follows:

$$\kappa_i = \begin{cases} 1 & \text{if } \epsilon_i \geq \epsilon_0 \\ \alpha(\epsilon_i/\epsilon_0)^{\nu} & \text{otherwise} \end{cases} \tag{7}$$

$$\kappa_i' = \frac{\kappa_i \times num_i}{\sum_{cl_k \in [A]_{-1}} (\kappa_k \times num_k)} \tag{8}$$

$$F_i \leftarrow F_i + \beta(\kappa_i' - F_i) \tag{9}$$

A crucial aspect of the XCS update process is the use of the relative accuracy κ' for calculating fitness. Specifically, by weighting the absolute accuracy κ according to the degree of polymerization, a rule is designed such that more general rules have a higher degree of conformity. Additionally, as shown in Eqs. 7, inaccurate rules are penalized with exponential decay. Consequently, κ' prioritizes the most accurate rule by assigning a higher fitness. This mechanism is essential because the validity of the learned rules cannot be guaranteed without improving the generality with respect to

accurate rules. Therefore, when comparing rules with equal absolute correctness, the more general rule with a higher degree of polymerization is retained in $[P]$. The use of κ' also introduces a selection pressure to reduce the number of rules in $[A]_{-1}$ by adaptively reflecting the dominance of the rules in $[A]_{-1}$ over F_i. Consequently, this is expected to improve the overlap of rules in the input.

4) DISCOVERY COMPONENT

The discovery component is responsible for evolving the classifiers by creating and deleting them using a GA. First, before applying the steady-state GA, XCS applies a subsumption on the rules within the previous action set $[A]_{-1}$. Subsumption aims to merge similar rules into more concise rules. To apply this operation, it is necessary to determine the subsumption relationship between the rule conditions. Consider two rules, cl_1 and cl_2, where cl_1 is determined to be more accurate. This determination is based on the threshold condition that the number of updates n must exceed θ_{sub}, indicating that sufficient updates have been made to ensure the reliability of the rule. Such a rule is referred to as a "reliably accurate" rule. Thus, if $cl_2.C \subseteq cl_1.C$, only cl_1 should be stored in $[P]$. Consequently, cl_2 is deleted from $[P]$ and cl_1 is increased to $num_1 \leftarrow num_1 + num_2$ to record the number of aggregate rules under the condition of $cl1.A = cl2.A$.

Finally, the steady-state GA is applied when the average number of steps since the last GA execution for each classifier in the action group $[A]_{-1}$ exceeds a threshold (θ_{GA}). This GA genetic operates by selecting two classifiers as parents using the relative probability of selection as the magnitude of the fitness of each classifier in $[A]_{-1}$. Crossover and mutation operations are then applied to these parents with a certain probability (χ, μ) to produce two children. These children are added to the classifier group, and if the populations exceeds the maximum N limit, they are removed based on their degree of fitness. It is important to note that while classifiers are generated and deleted by this discovery component, it searches for the appropriate range of generalization in the classifier conditions. Applying the GA to $[A]_{-1}$ is important because more general rules tend to appear in $[A]_{-1}$ more frequently. Consequently, the GA has a higher probability of generating rules under similar conditions. Therefore, the degree of polymerization tends to increase after inclusion. Additionally, the inclusion operation is only applied to accurate rules that can be trusted. Consequently, for general but imprecise rules ($\epsilon_n \geq \epsilon_0$), the degree of polymerization does not increase. This mechanism effectively prevents overgeneralization.

5) REAL VALUE EXTENSION (XCSI)

XCSI [42] has been primarily used in data mining for real-valued data to apply XCS to real-valued problems with LU representation, which is based on lower and upper values. The condition part of the classifier (C) has the same number of dimensions as the input state x, and the condition part C_i in each dimension i comprises a real-value interval $[l_i, u_i)$. Therefore, the condition for a classifier to match state s is $l_i \leq x_i < u_i$ in all dimensions. If no classifier matches the input state x, the conditional part of the classifier generated by the covering operation is determined using Eq. 10, where rand (s_0) is a random number capped by the parameter s_0. Then, for a real-valued range of input states $[x_{min}, x_{max}]$, l_i is set to x_{min} if $l_i > x_{min}$ and u_i is set to x_{max} if $ui > x_{max}$.

$$\begin{cases} l_i = s_i - rand(s_0) \\ u_i = s_i - rand(s_0) \end{cases} \tag{10}$$

Subsumption in the real-valued representation is performed based on the inclusion relation of the real-valued intervals of the classifiers. Specifically, for real-value intervals (l_i, u_i) and (l_i', u_i') of classifiers X and Y, X is assumed to be more general than Y if $l_i \leq l_i'$ and $u_i' \leq u_i$ for all dimensions, and the subsumption operation is executed.

The XCSI used in this study allows for real-valued conditions and actions, making it suitable for complex air-traffic scenarios. For instance, when controlling aircraft speed based on proximity and congestion levels, XCSI can generalize the rules that apply to various traffic conditions while ensuring interpretability.

IV. LEARNING PERFORMANCE OF XCSI

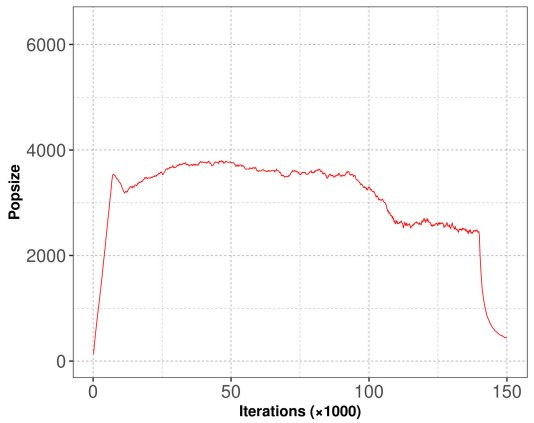
A. EXPERIMENTAL SETTING

The XCSI algorithm was implemented in C++ based on the open-source code *XCS Java* [47]. The experiments involved training XCSI on a dataset for a specified number of training cycles and then evaluating its performance on a separate evaluation dataset. During training, actions were selected randomly. Additionally, performance and discovery components are not applied to the evaluation data.

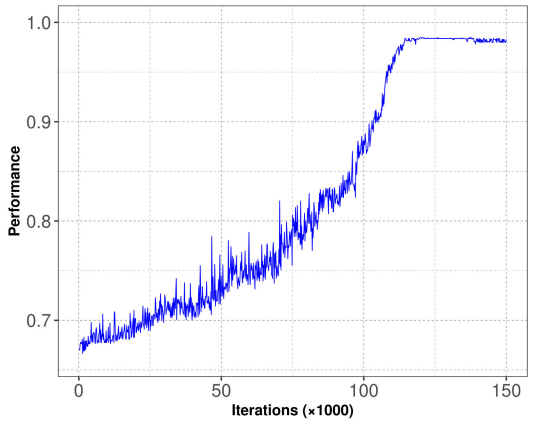
To ensure reliable learning, we evaluated the correct response rate for each dataset using crossvalidation ($k = 3$), similar to a previous study [48]. The training employed early stopping based on the crossvalidation accuracy and dropout techniques were included to mitigate overfitting. Specifically, the algorithm was trained for up to 150, 000 cycles, and its performance was regularly evaluated on a validation set to monitor its performance and prevent overfitting by halting training once the accuracy plateaued. After training, the model's performance was assessed on the evaluation dataset using an ϵ-greedy action selection strategy. The accuracy of the selected actions was established as the evaluation metric.

The parameters of XCSI were configured according to recommended values in [47]: $\nu = 5$, $\alpha = 0.1$, $\beta = 0.2$, $\epsilon_0 = 0.001$, $\theta_{mna} = 2$, $\theta_{GA} = 50$, $p_\chi = 0.8$, $p_M = 0.04$, $P_\# = 0.33$. Subsumption was applied with a threshold of $\theta_{sub} = 50$, and the maximum number of classifiers was limited to $N = 6, 400$.

Additionally, after 130,000 training cycles, $p_\chi = 0.8$ and $p_M = 0.04$ were set to 0. This adjustment helped maintain accuracy while reducing the number of rules, as both accuracy and the number of rules had sufficiently converged by this point.



(a) Population Size



(b) Accuracy

FIGURE 5. Line plot of population size and accuracy.

B. ACCURACY AND POPULATION SIZE

Fig. 5 shows the line plot of accuracy and population size. Evidently, XCSI significantly outperformed the decision-tree model [28] in both the number of generated rules and accuracy achieved during training. As the number of training iterations increased, XCSI gradually produced more rules, ultimately reaching 433 rules, while simultaneously improving its accuracy to 0.96. This incremental growth in both rule count and accuracy demonstrates the ability of XCSI to adapt to environmental complexities. By contrast, the decision tree generated a relatively small and static set of eight rules with a moderate accuracy of 0.70, which plateaued as the training progressed. The flexibility and adaptability of XCSI, stemming from its evolutionary approach, allowed it to outperform the decision-tree model, particularly in scenarios requiring nuanced decision-making.

V. SPEED-CONTROL EFFECTIVENESS OF XCSI IN TRAFFIC SIMULATOR

A. EXPERIMENTAL SETTING

In the experiments, we used an XCSI-trained speed-control model in a traffic simulator. That is, the environment shown in Fig. 4 is altered from the test data (Section IV) to the CA-based air-traffic model. Initially, aircraft characteristics x, such as distance to the preceding aircraft and the congestion level, were input into the XCSI model at the 150-NM point from the airport. The model then predicted the appropriate speed-control action based on the selection probability (Eq. 3), which was employed in the simulator to adjust the aircraft's velocity. Simultaneously, the match set $[M]$, prediction array $P[x, A]$, and action set $[A]$ were stored to analyze the rules contributing to the speed-control action. This loop was repeated for all inbound flights coming southwest of Tokyo International Airport, and the results were averaged to assess the overall effectiveness of the speed-control rules. Thus, this experiment validated the effectiveness of speed control by the trained model.

TABLE 2. Cases used for model comparisons.

Case	Definition
Original	No speed control
Opt	Speed control with non-dominated solution
DT_bin	Model that learns "with/without" speed control using decision tree Extent of speed control : {17kt} (when speed control is implemented) Tree depth : {3 layers} [28]
XCS_bin	Model that learns "with/without" speed control using XCSI Extent of speed control: {17kt} (when speed control is implemented)

Table 2 describes the four comparison cases used in this study to evaluate the effectiveness of the various speed-control strategies. The ''Original'' case served as the baseline, wherein no speed control was employed. The ''Opt'' case implemented speed control based on the non-dominated solutions derived from the optimization process, whereas the ''DT_bin'' case employed a decision-tree model that learns whether to apply speed control or not using binary classification. In this case, a speed reduction of 17 kt was applied when the speed control was triggered, and the decision tree featured a depth of three layers, based on the model proposed by [28]. Finally, the ''XCS_bin'' case employed XCSI to learn the binary decision of whether to apply speed control, with the same speed reduction of 17 kt under the application of the control. These models allowed comparing the effectiveness of the various speed-control strategies under the same conditions.

B. IMPROVEMENT IN THE OBJECTIVE FUNCTION SPACES

The comparison of the objective function spaces, illustrated in Fig. 6, demonstrates the superiority of XCS_bin over both the original scenario without speed control and the

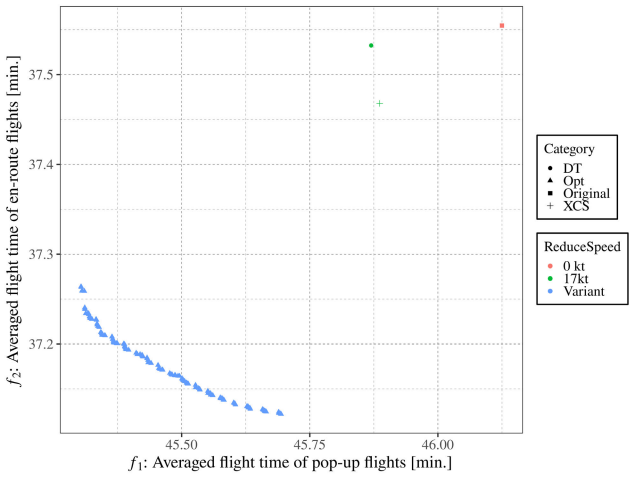


FIGURE 6. Objective function space.

rules based on decision trees. The analysis plots the average flight times for en route inflow and departing aircraft, revealing that the solution derived from XCSI is consistently closer to the non-dominated solutions, as indicated by the locations of the plotted points. Specifically, the XCSI-based rules achieved more effective flight time reductions while maintaining a better balance between the two objectives. In particular, the XCSI-derived solutions are positioned lower in the objective function space than those without speed control, signifying an improvement in overall efficiency. This outcome highlights the ability of XCSI to learn and adapt to the complex dynamics of air traffic, thereby enhancing the system performance and contributing to a more optimal trade-off between the flight times of the en route and departing aircraft. Consequently, XCSI is more capable of improving the overall flight time by fine-tuning the speed controls in a manner that closely aligns with the desired objectives.

TABLE 3. Comparison of key operational indicators.

	Original	DT_bin	XCS_bin
Total Flight Time [s]	1280832	1279427	1278031
Total Fuel Consumption [kg]	837881.2	836404.4	836231.1

C. IMPROVEMENT IN KEY OPERATIONAL INDICATORS

Table 3 demonstrates the operational benefits of applying XCSI-based speed control (XCS_bin) over both decision-tree-based control (DT_bin) and the baseline scenario without speed control (Original). Specifically, the total flight time was reduced by approximately 1,801 s when using XCSI, while the fuel consumption was decreased by 1,650.1 kg compared to the original case. This improvement suggests that XCS_bin effectively streamlines arrival operations by optimizing speed adjustments, leading to enhanced efficiency in air traffic flow. Additionally, compared to DT_bin, the XCSI model further reduced both flight time and fuel consumption, indicating that its rule-based approach is more effective in balancing traffic efficiency and energy usage. This advantage stems from the ability of XCSI to derive interpretable and adaptive rules that closely align with multiobjective optimal solutions, ensuring both precision in speed control decisions and their generalizability across varying traffic conditions.

The observed reduction in fuel consumption also highlights a significant economic benefit, as lower fuel usage directly translates into cost savings for airlines. Based on IATA's reported fuel price for the week ending January 24, 2025, at $773.27 per ton [49], the reduction of 1,650.1 kg (1.65 tons) of fuel the XCSI-based approach results in an estimated cost savings of approximately $1,276 per operational cycle. Realizing this saving daily over the year would lead to an annual economic benefit of approximately $466,000 per flight operation. This demonstrates the potential of XCSI-based decision support for substantial cost reductions when applied at scale, making it a valuable strategy for airlines aiming to optimize operational efficiency and reduce fuel expenses.

TABLE 4. Confusion Matrix of DT_bin.

		Prediction Value	
		SpeedReduction	NominalSpeed
True Value	SpeedReduction	69	68
	NominalSpeed	59	235

TABLE 5. Confusion matrix of XCS_bin.

		Prediction Value	
		SpeedReduction	NominalSpeed
True Value	SpeedReduction	131	6
	NominalSpeed	8	286

TABLE 6. Performance summary.

	DT_bin	XCS_bin
Accuracy	0.705	0.968
Precision	0.539	0.942
Recall	0.503	0.956
F-score	0.520	0.949
f_1 Improvement	0.255	0.239
f_2 Improvement	0.022	0.087

D. CONFUSION MATRIX

The performance metrics listed in Tables 4 and 6, comparing the application of rules derived from the XCSI (XCS_bin) and decision-tree model (DT_bin) using a confusion matrix, highlight the superior accuracy of XCS_bin in the simulator environment. Additionally, the precision and recall scores indicate that XCSI significantly outperformed the decision-tree model. Specifically, XCS_bin achieved a precision of 0.942 and recall of 0.956, whereas the decision tree demonstrated a precision of 0.539 and recall of 0.503. In the

context of ATM, high precision and recall are vital for ensuring safety and efficiency, as they minimize the risk of incorrect or missed interventions that could lead to congestion or delays. Because the LCS pre-generates rules aligned with multiobjective optimal solutions during the training phase, a high precision, recall, and F-score indicate that the generated rules closely approximate the optimal speed control decisions. This demonstrates the effectiveness of XCSI in deriving decision rules that support efficient air traffic management, ensuring that the system consistently provides reliable and actionable guidance to ATCos. Thus, these results demonstrate that XCSI not only enhances the accuracy of rule application but also the overall system performance in dynamic ATM scenarios.

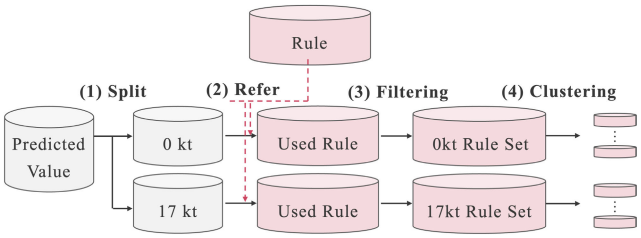


FIGURE 7. Framework of the rule-analysis process. (1) Splitting the data into prediction classes for 142 flights. (2) Rules used for each prediction class collected by referring to a set of rules. (3) Collection process of the rules used in the prediction class through filtering using $F_i \geq 0.6$ and $p_i = 1000$. (4) Clustering each rule set via hierarchical clustering.

VI. ANALYSIS OF OBTAINED SPEED-CONTROL RULE

In this section, we extract and analyze the rules contributing to the decision to control the speeds of 431 flights. Specifically, we analyzed rules with a high predicted reward p_i and fitness F_i to identify situations that increase the probability of selecting each speed-control value. This analysis identifies the speed control should be consciously applied by ATCos in specific situations, as well as the consequences of not applying speed control.

A. METHOD

Based on the insights gained from the learning-performance analysis presented in Section V-D, we focused on the deceleration rules (17kt Rule Set) as they have demonstrated significant effectiveness in managing aircraft spacing and minimizing congestion near TMAs. Prioritizing deceleration helps ensure smoother traffic flow and minimizes the risk of sudden adjustments during the approach phase.

In the next step, hierarchical clustering was applied to each rule set (Step 4 in Fig. 7). Based on the Euclidean distance, each dataset was clustered according to the upper (u_i) and lower (l_i) bounds of the rules using the Ward method. In this step, the binary values *FlightCategory* and *IsCluster1* are converted into upper and lower bounds as follows:

$$l_i = \begin{cases} 0 & \text{if } FlightCategory = 0 \text{ or\#} \\ 1 & \text{otherwise} \end{cases} \tag{11}$$

$$u_i = \begin{cases} 0 & \text{if } FlightCategory = 0 \\ 1 & \text{otherwise} \end{cases} \tag{12}$$

$$l_i = \begin{cases} 0 & \text{if } IsCluster1 = 0 \text{ or\#} \\ 1 & \text{otherwise} \end{cases} \tag{13}$$

$$u_i = \begin{cases} 0 & \text{if } IsCluster1 = 0 \\ 1 & \text{otherwise} \end{cases} \tag{14}$$

Hierarchical clustering was performed on the extracted rules using the *NbClust* package [50] in *R*. The optimal number of clusters was determined by considering all available indices in the *NbClust* package, including silhouette analysis, the elbow method, and gap statistics. From each cluster, the rule with the highest fitness F_i and frequency of use was selected as the representative rule, and the conditional components of the representative values were then compared to identify patterns and differences between the clusters.

To further understand the influence of these representative rules, we analyzed their contribution to the selection probability. Specifically, we evaluated the proportion of the total selection probability attributed to aircraft matching the conditions of each representative rule (see Eq. 3). Finally, we calculated the key performance indicators (KPIs) for aircraft affected by these rules, including those arriving within a 600-s window before and after the arrival of these aircraft. These KPIs, which included the number of spacing adjustments, total flight time, total fuel consumption, and flight time within TACA, were compared to the baseline scenario (without speed control) to assess the impact of the rules on air-traffic efficiency and flow.

B. HEATMAP OF SPEED REDUCTION RULE

The hierarchical clustering analysis applied to the speed-control rules, visualized in the heatmap in Fig. 8, reveals significant patterns and relationships. The color intensity indicates the strength of each rule's normalized conditions. Those with higher normalized conditions (closer to 1) are more dominant in their respective clusters, indicating a stronger influence on speed-control decisions. The heat map clearly delineates four main clusters (G1, G2, G3, and G4), each grouping rules with similar operational characteristics.

The relationships between the rules in the four groups and specific traffic conditions are as follows:

- Cluster G3 is characterized by rules where *FlightCategory_l* $= 1$, indicating a strong association with flights (excluding pop-up departures). Furthermore, nearly all rules in G3 feature *IsCluster1_l* $= 0$ and *IsCluster1_u* $= 1$, suggesting that sector-common rules were successfully extracted.
- Clusters G4 and G2 are distinguished by the combination of *IsCluster1_l* $= 1$ or *IsCluster1_u* $= 0$ with *DtAC_1_u*, suggesting a specialized rule set tailored for scenarios involving a specific incoming direction (Cluster 1 or 2) and distance to the preceding aircraft. These rules are particularly beneficial when aircraft merge into busy traffic flows from en route sectors,

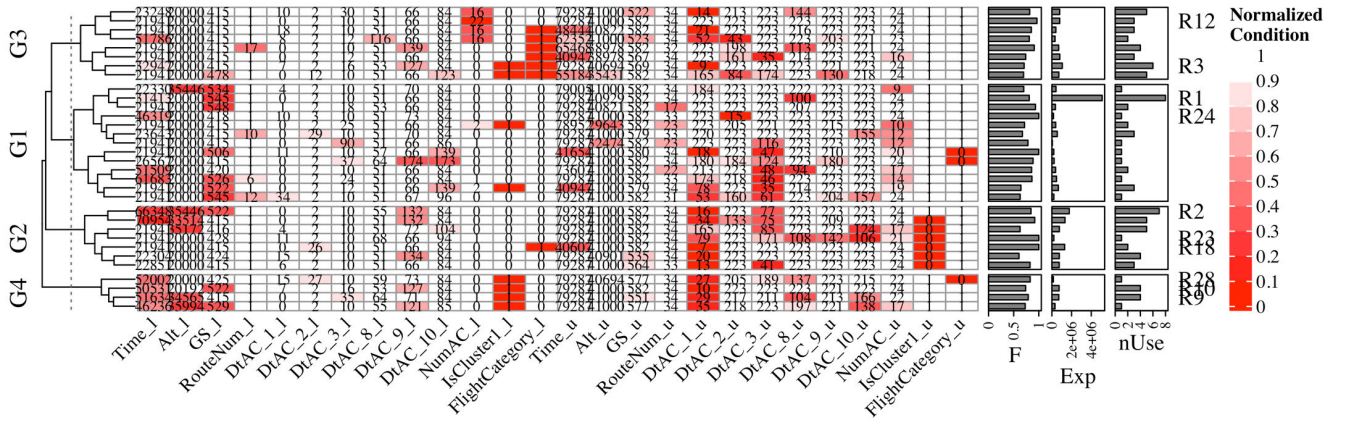


FIGURE 8. Heatmap of speed-control rule application across different flight conditions. The color gradient reflects the strength of each rule's condition, revealing distinct patterns associated with different clusters and traffic conditions. This visualization provides valuable insights into the operational conditions that trigger specific control strategies.

as they help maintain safe separation distances and avoid last-minute spacing adjustments closer to the TMA.

- With respect to G1, the rules tend to be activated under conditions involving specific combinations of ground speed (*GS_l*) and time (*Time_l*).

These distinct clusters demonstrate that certain rules are consistently applied together, indicating their effectiveness in similar scenarios. These findings highlight the adaptability of XCSI, as the extracted rules reflect the system's ability to customize control strategies for different operational scenarios.

Moreover, the heatmap reveals variations in rule usage across different scenarios, with some rules used extensively (e.g., R1, R2, and R3) and others used less frequently. However, even infrequently used rules can significantly increase the probability of speed control if they have a high fitness (F) score, indicating its effectiveness under specific and often highly specialized conditions (e.g., R23). This clustering not only illustrates the system's ability to adapt and optimize rule sets for different conditions but also provides insights into which rules are most critical for successful ATM. The hierarchical structure further reveals sub-clusters within these groups, suggesting nuanced variations in rule applications that could be explored for developing more refined control strategies in the future.

C. CONTRIBUTION TO SELECTION PROBABILITY

Analysis of the proportion of selection probability for the representative rules (R1, R24, R2, R18, R23, R3, R12, R9, R10, and R28) revealed a clear relationship between the rule IDs and their frequency of use in the simulation. As reflected in the box plot shown in Fig. 9, rules with lower IDs, such as R1 and R2, tend to exhibit higher selection probabilities, indicating their frequent application and central role in the decision-making process. These rules are pivotal in guiding operational decisions because they are designed to address more commonly encountered scenarios within the simulation. Conversely, rules with higher IDs, such as R24 and R28, are used less frequently, owing to their specialized nature. Despite their lower usage frequency, these rules exhibit a strong selection probability in specific situations, emphasizing their critical role in addressing specific and less common scenarios. These findings underscore the importance of both frequently used general-purpose and more specialized rules for addressing unique situations within the simulation.

D. VISUALIZING REPRESENTATIVE RULE ACTIVATION WITH BITMAPS

To provide a better understanding of the operational characteristics and implications of the extracted speed-control rules, Figs. 10–13 employ bitmaps to present detailed visualizations of the specific parameters of each rule and their potential impacts on ATM. The following sections describe these figures and explain how each rule (G1(R1), G2(R2), G3(R3), and G4(R9)) contributes to safe and efficient airspace management under various scenarios.

Analysis of G1(R1) reveals that this rule is primarily concerned with the early deceleration of aircrafts operating at high altitudes and speeds. Specifically, it mandates speed control for aircraft with ground speeds exceeding 545 kt. The rule does not enforce a strict timing for initiating speed control, allowing for flexibility the response to varying traffic conditions. Furthermore, the requirement for speed control is triggered when the distance to the eighth-preceding aircraft falls within 100 NM, a relatively long distance that allows for gradual deceleration. This approach is likely to promote smoother navigation and enhance safety, particularly in congested airspaces. The early deceleration mandated by G1(R1) contributes to maintaining a safe separation between aircraft, thereby reducing the collision risks.

In contrast to G1(R1), G2(R2) emphasizes the importance of precise timing in speed-control operations. The rule mandates speed control intervention at a specific time

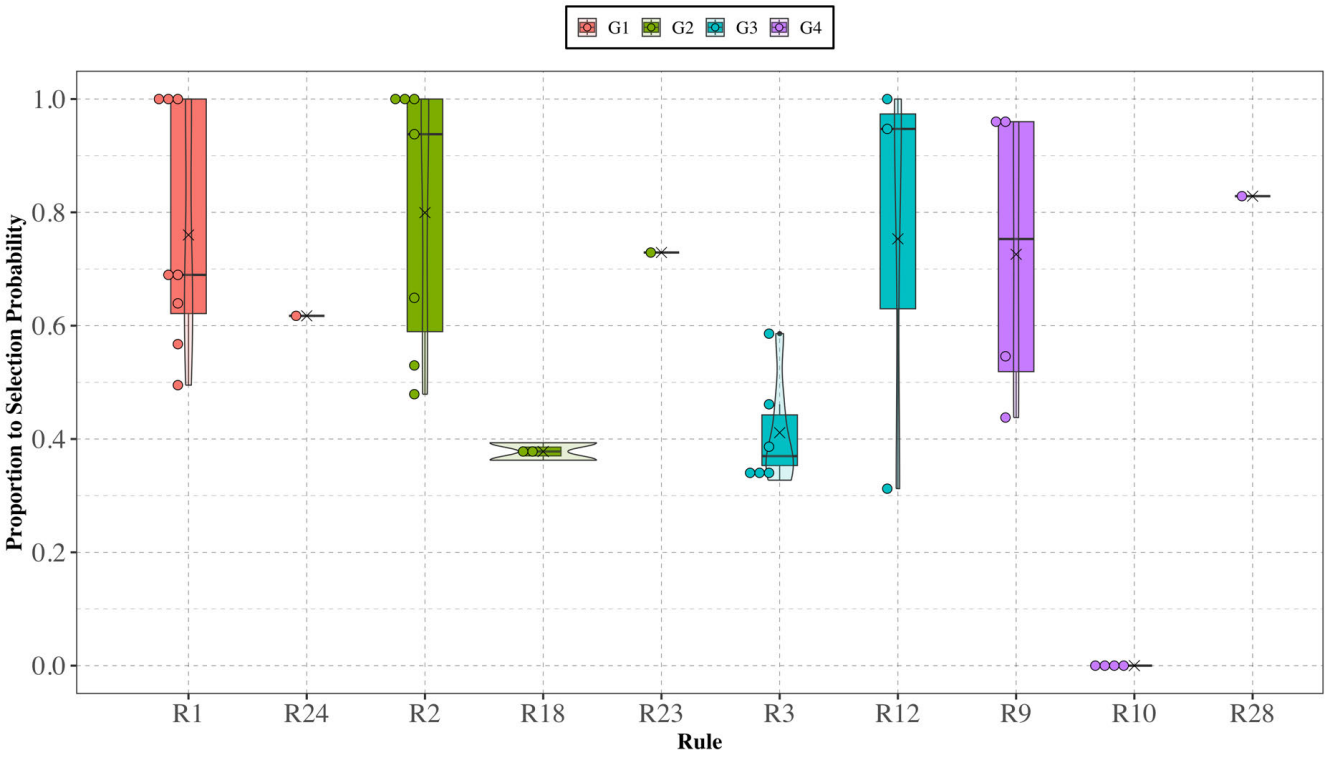


FIGURE 9. Boxplot illustrating the proportion of each speed-control rule's contribution to decision-making within the simulation, based on hierarchical clustering applied to the actual rules used during the simulation. Each boxplot represents a rule selected from clusters generated through hierarchical clustering. The representative rules (R1, R24, R2, R18, R23, R3, R12, R9, R10, R28) were selected based on their significance in operational scenarios. The x-axis denotes the number of rules, whereas the y-axis indicates the selection probability attributed to each rule, with the "×" symbol representing the mean selection probability. The distribution and spread of these proportions highlight their relative importance and frequency of use in simulated ATM scenarios.

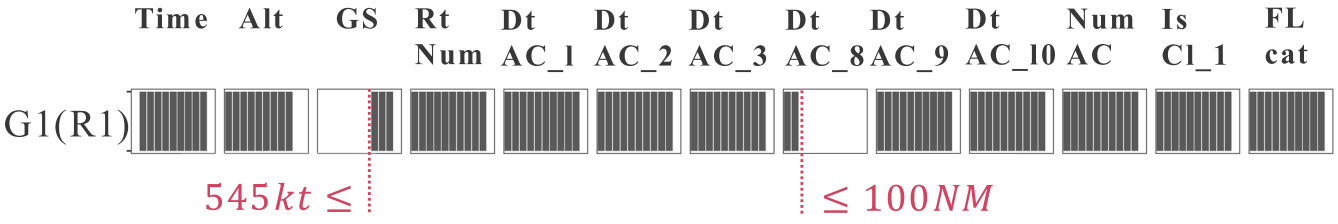


FIGURE 10. Activation bitmap for Rule G1(R1), which classifies individual aircraft under speed control based on their operational characteristics. The bitmap allows for a detailed comparison of how these rules are distributed across various flight scenarios and their underlying similarities and differences.

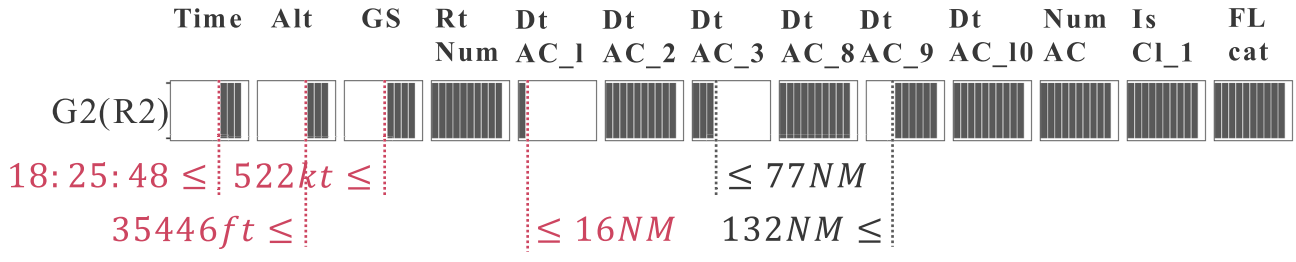


FIGURE 11. Activation bitmap for Rule G2(R2), which classifies individual aircraft under speed control based on their operational characteristics. The bitmap allows for a detailed comparison of how these rules are distributed across various flight scenarios and their underlying similarities and differences.

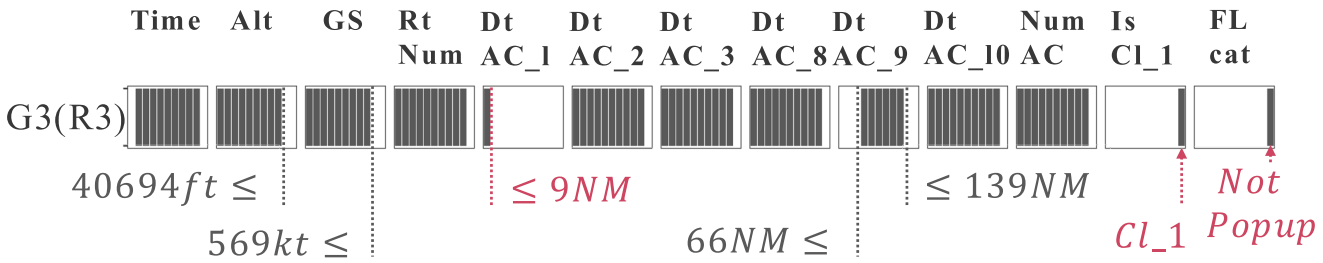


FIGURE 12. Activation bitmap for Rule G3(R3), which classifies individual aircraft under speed control based on their operational characteristics. The bitmap allows for a detailed comparison of how these rules are distributed across various flight scenarios and their underlying similarities and differences.

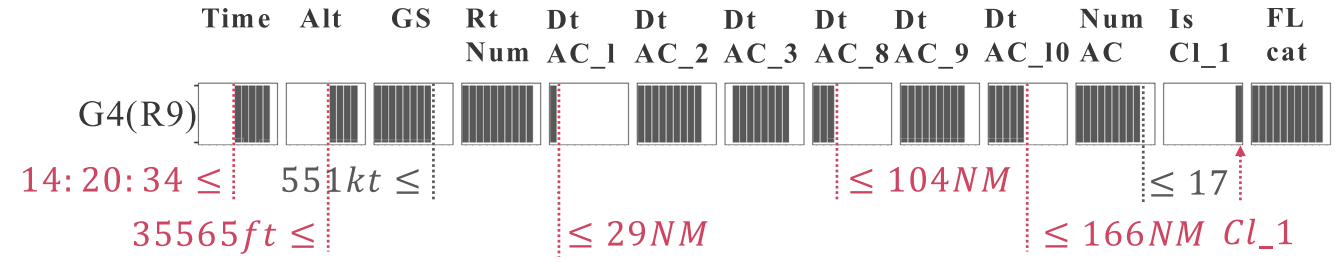


FIGURE 13. Activation bitmap for Rule G4(R9), which classifies individual aircraft under speed control based on their operational characteristics. The bitmap allows for a detailed comparison of how these rules are distributed across various flight scenarios and their underlying similarities and differences.

(18:25:48), suggesting its relevance in scenarios with time-critical factors. Although the altitude requirement is set at 35,446 ft or higher, similar to G1(R1), the distance to the preceding aircraft that triggers speed control is notably shorter, at just 16 NM. This close proximity necessitates rapid deceleration, which is crucial for managing situations wherein aircraft converge quickly. G2(R2) is designed to address scenarios requiring urgent speed adjustments to maintain safe separation and prevent potential conflicts in dense airspaces.

Rule G3(R3) introduces a different operational scenario, focusing on the need for rapid deceleration at high altitudes and speeds. Unlike G2(R2), G3(R3) does not impose strict timing requirements, allowing for more flexibility in the implementing speed-control actions. However, similar to G2(R2), the distance at which speed control must be initiated is very short, at just 9 NM, highlighting the need for quick responses to maintain a safe separation in this high speed scenario. This rule is particularly relevant in high-altitude airspaces, where aircraft operate at high speeds, necessitating swift deceleration to manage traffic effectively and ensure safety.

Rule G4(R9) combines the elements of both time-specific and flexible operational strategies. It mandates speed control at the precise time of 14:20:34 and applies to aircraft operating above 35,565 ft with ground speeds exceeding 551 kt. Unlike the other rules, G4(R9) allows deceleration at a relatively long distance of 29 NM from the preceding aircraft. This extended distance offers greater flexibility for implementing speed-control measures, allowing earlier adjustments that contribute to smoother traffic flow and enhanced safety. The focus of the rule on timing and distance suggests a balanced approach that supports efficient airspace management while maintaining sufficient safety margins.

E. KPI IMPROVEMENT

Fig. 14 illustrates the KPIs comparing the ATM performance under rules R1, R2, R3, and R9. These indicators include the number of spacing adjustments (CAadj), duration (total flight time), fuel consumption, and time spent in TACA (TACA). These metrics were evaluated based on the differences between the simulations with rule-based speed control and the baseline (original) simulations. The findings reveal the different impacts of each rule on ATM and flight efficiency.

The number of spacing adjustments reveals a significant variance between the rules. Specifically, R2 and R3 exhibit an average decrease in adjustments by approximately 5 counts, suggesting effective spacing without frequent intervention, which can be attributed to its preventive speed-control strategy. Although R1 and R9 also show variations, R1 maintains a closer consistency with R9, suggesting its suitability for scenarios requiring less drastic spacing adjustments.

The total flight time is minimally altered under R3, with an average decrease of only 3 s, suggesting that its control mechanisms did not significantly extend flight duration. By contrast, R1 and R9 demonstrate wider variances in flight time, with R9 leading to reductions of up to 12 s on average. Finally, R2 mirrors the pattern of R3 but with an even smaller impact on time, reducing it by an average of 5 s in several

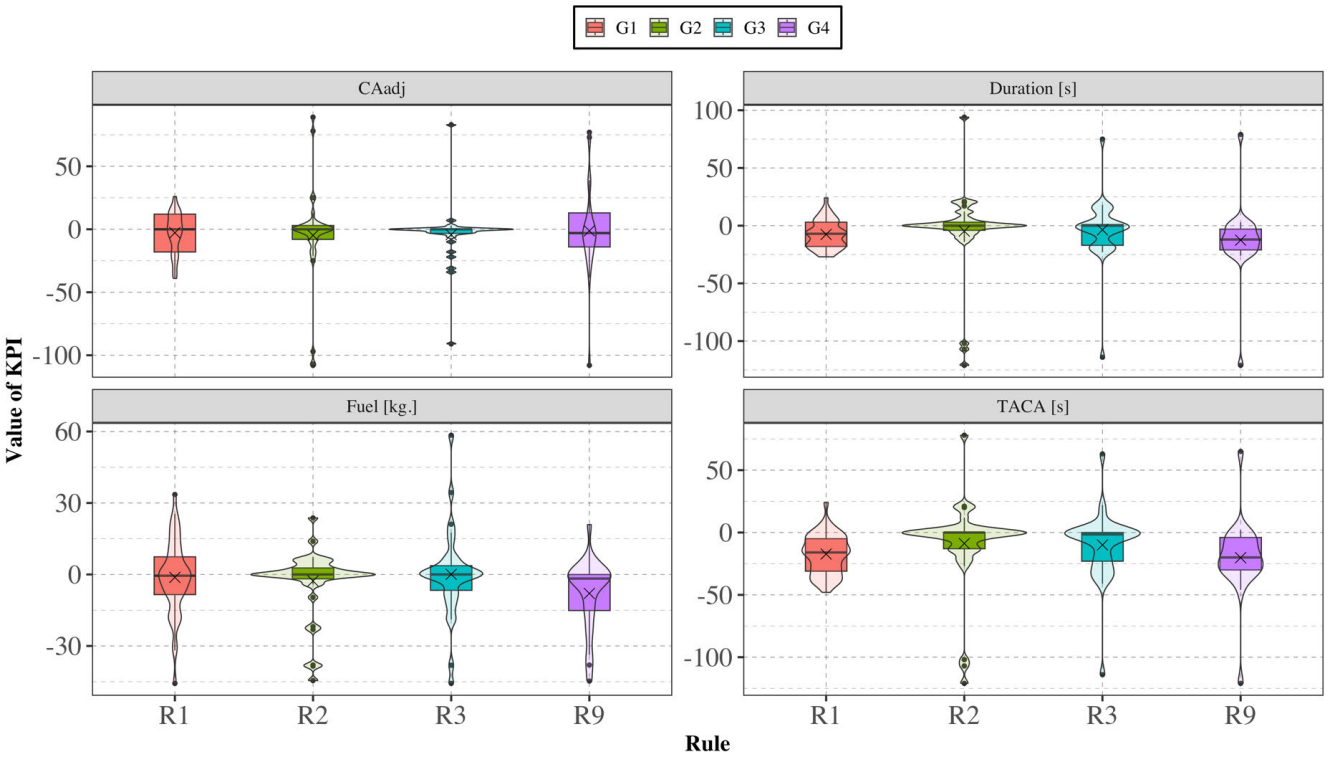


FIGURE 14. Impact of Speed Control Rules on KPIs Evaluation: This boxplot compares the KPIs for aircraft under different speed-control rules (R1, R2, R3, and R9). The "×" symbol indicates the mean value. Each KPI reflects the difference between rule-based and original (baseline) simulations, where the original simulation is subtracted to compute the differential values. CAadj denotes the number of spacing adjustments made under rule-based simulation, Duration [s] is the total difference in flight time, Fuel [kg.] is the difference in fuel consumption between the rule-based and baseline simulations, and TACA [s] is the flight time through TACA, calculated from the perspective of the downstream en route airspace.

cases, suggesting streamlined operations within controlled airspace.

Fuel consumption also varies across the rules. R3 shows the smallest change, with a negligible average increase of 0.07 kg, indicating that while effective for managing spacing and reducing flight time, it may not be optimizing speed adjustments for fuel efficiency. However, R9 decreases the fuel consumption by upto 7.96 kg on average compared with the baseline, potentially owing to an optimal balance between speed control and fuel efficiency. The fuel consumption under R1 exhibits considerable fluctuations, suggesting inconsistent fuel efficiency, whereas R2 typically reduces fuel consumption by 3 kg on average, demonstrating its ability to enhance the fuel economy through more consistent speed management.

The time spent in TACA indicates that R1 and R9 can reduce the TACA time by up to 20 s on average, suggesting effective management of aircraft speeds when approaching busy terminals. R2 and R3 show minimal changes, maintaining their stability and efficient handling in the approach areas.

This KPI-based evaluation of these rules suggests that R1 and R9 offer the most stable management across different metrics, leading to minimal disruptions in flight operations. Although R3 is effective in certain high-density traffic scenarios, it introduces variability that can potentially affect operational consistency. By contrast, R2 reduces both flight times and fuel consumption; therefore, it can be beneficial in scenarios balancing fuel cost and operational delays. These insights are instrumental in tailoring rule applications to specific air-traffic conditions for optimizing both safety and efficiency.

VII. DISCUSSION

This study aimed to develop a novel framework for arrival management using XCSI to enhance delay reduction near destination airports through adaptive speed control and rule-based decision-making. The framework was designed to distill complex speed control strategies into interpretable and actionable rules, facilitating the analysis of speed control practices in En Route AMAN by accounting for factors such as relative spacing and traffic flow. This research aimed to validate the learning performance of the XCSI algorithm and assess the practical utility of the extracted rules to support ATM decision-making. The key research question guiding this study was: *"Can complex speed control strategies be effectively simplified into concise and interpretable rules using XCSI while maintaining adaptability to varying traffic conditions?"*

The results presented in this paper provide a compelling answer to this question. XCSI successfully extracted interpretable and effective speed control rules, capturing distinct control patterns across various areas and conditions, such as T25 and T14. These patterns include rules for adjusting speed based on aircraft spacing and traffic density, resulting in a high contribution to selection probability (see Fig. 9) and a notable reduction in average flight times by 10–20 s (see Fig. 14). For instance, a rule involving speed reduction when the distance to the preceding aircraft was less than 9 NM effectively managed congestion in the T25 region (see R3 in Fig. 8 and Fig. 12), similar to a rule obtained previously through decision-tree analysis [28]. Similarly, in T14, rules that focused on maintaining relative spacing with the first and third preceding aircraft proved beneficial for reducing delays during peak hours (see G2 in Fig. 8). Furthermore, the XCSI approach identified common rules applicable across both T25 and T14, demonstrating the ability to generalize speed control strategies across different airspace sectors (see G3 in Fig. 8). These extracted rules can be seamlessly incorporated into ATCos' decision-support systems to enhance proactive and informed speed management in en route airspace. Such integration could potentially reduce the need for vectoring and holding delays, leading to smoother traffic flow and improved efficiency. Specifically, we envision a decision-support system that first indicates whether deceleration is required, allowing controllers to quickly assess the need for speed adjustments. If controllers require further information or validation, the system can then allow them to review the specific rules underlying the deceleration suggestion. This two-step approach balances immediate guidance with the transparency and explainability of the underlying decision logic, empowering ATCos to optimize speed adjustments according to local traffic conditions.

This approach offers a novel application of XCSI in ATM, enabling the derivation of generalized rules from complex control decisions. Unlike traditional black-box models, XCSI allows for the interpretation of rules, which facilitates understanding and practical application. The algorithm generates rules that are both consistent with and generalizable to optimal speed control solutions, providing insights into the strategies suitable for different control scenarios. Moreover, XCSI can evaluate the contribution of these rules according to specific traffic conditions, enabling a nuanced understanding of their impact. While machine learning methods such as ensemble models or deep learning could potentially improve accuracy over decision trees and may yield results comparable to XCSI, these methods often lack interpretability, making it challenging to understand the reasoning behind their decisions. For instance, deep neural networks (DNNs) function as black-box models, offering no direct insight into their decision-making process. In contrast, XCSI provides transparent and interpretable decision-making without compromising quantitative rigor or objectivity. This interpretability is crucial in ATM, where understanding the rationale behind speed control decisions is essential for building trust and ensuring safe and explainable operations. Although a detailed quantitative comparison with these other models is beyond the scope of this study, it remains an important area for future research. Another key advantage of XCSI is its ability to extract multiple rules that account for varying traffic situations, even when the same speed control action is applied. By using hierarchical clustering to classify rules into patterns, the approach can present a range of speed control rules tailored to different scenarios. This supports intuitive decision-making by ATCos, allowing for adaptive control within realistic operational constraints.

Compared to our previous study [28], which focused on decision-tree analysis, the current approach using XCSI offers complementary strengths and weaknesses. Decision trees excel at extracting hierarchical structures, allowing ATCos to intuitively grasp broad speed control strategies through their visual representation. However, this hierarchical structure can impose limitations, as decision trees rely on fixed structures with "AND" conditions that may restrict flexibility in dynamic scenarios. By contrast, XCSI avoids these constraints by enabling flexible rule construction, accommodating diverse scenarios without hierarchical limitations. This flexibility allows XCSI to extract both common and unique rules for different airspace sectors, which can serve as a dynamic knowledge-sharing database. However, this adaptability comes at the expense of the intuitive visualization offered by decision trees. While decision trees provide a coarse-grained, high-level understanding of speed control strategies suitable for initial analysis, XCSI specializes in generating precise, adaptable rules for real-time applications. The proposed LCS approach demonstrated superior performance, achieving closer alignment with reference solutions (Fig. 6), higher precision during training (see Fig. 5), and significantly better recall in simulations (see Fig. 6), highlighting its effectiveness for detailed en route traffic management.

Unlike previous research focused on optimization-based methods [15], [16], [17], [18], [19], [20], [21], this study emphasizes rule extraction, which can facilitate knowledge sharing among controllers. While optimization approaches may offer more precise delay reductions in specific scenarios, this rule-based framework enhances situational adaptability by supporting diverse control patterns. Thus, this study lays the foundation for future developments in adaptive and context-sensitive traffic management. From a data exploration perspective, this study stands out as one of the few real-world data mining cases within the context of LCS [35], [39], [40], [41]. Although there are no substantial improvements to XCSI itself, this research demonstrates a novel application of transforming binary features and applying clustering techniques to analyze trends within clusters in detail. Such comprehensive analysis of cluster-specific tendencies has been largely absent in prior studies.

To facilitate real-world implementation, it is necessary to refine the algorithm and validate the effectiveness of the rules in real-world scenarios. The proposed LCS framework

generates rules in advance during the learning phase by aligning them with multiobjective optimal solutions; therefore, the computational cost during operational deployment is minimal, as only precomputed rules are used. However, the features employed in this study, such as DtAC_i and NumAC, must be replaced with equivalent features that can be extracted in real-world operations. Converting features obtained from actual systems into the ones used in this study is relatively straightforward, and this step is essential for the operational deployment of the proposed system. Regarding dynamic factors, the proposed approach already considers weather conditions on specific dates and aircraft performance using the BADA model. To further incorporate dynamic elements, such as varying weather conditions and aircraft performance characteristics, we can include them as additional features in the model and/or expand the training dataset by performing multiobjective optimization across multiple dates with diverse weather conditions and aircraft types. This approach can enhance the adaptability of our system to dynamic operational conditions. Following this, conducting Human-in-the-Loop (HITL) simulations [29] will be crucial for evaluating the system's operational feasibility and validating its effectiveness in a realistic ATC environment, allowing for further refinements based on human feedback and real-world operational constraints.

In terms of scalability, this method is particularly suitable for airspaces transitioning from flow-based management to time-based management, specifically where ''metering'' (interval adjustment) begins. This is because the feature values utilized are based on relative intervals and traffic flow. However, this method has a potential limitation in adapting to temporal variations, changes in weather conditions, and high-density traffic scenarios. For example, while the XCSI in this study was applied to RJTT's predominant arrival flow, one of the busiest air sectors globally, further validation is needed in even more congested airspace, such as major hub airports with traffic converging from multiple directions. The challenges in such high-density scenarios include increased computational costs for rule extraction and potentially reduced rule generalization accuracy. Therefore, future work should expand the training dataset by incorporating diverse traffic patterns and weather conditions, and validate the rules in high-density scenarios to assess and improve the scalability and robustness of the approach. Furthermore, while this study has pioneered a novel application of single-step LCS, the proposed approach must be extended to multi-step control, where aircraft actions can be continuously adjusted over time. For urban air mobility applications, where real-time traffic coordination and dynamic route adjustments will be crucial in dense and complex urban environments, future work could explore multi-step XCS approaches to enable more adaptive and responsive control strategies.

VIII. CONCLUSION

This paper introduced a novel framework for en route arrival management using XCSI, aimed at reducing delays near destination airports through adaptive speed control and interpretable rules. The framework utilized a virtual design database generated from multiobjective optimization integrated with a CA-based air-traffic model, focusing on inbound flights to Tokyo International Airport. This database included binary target variables for speed control decisions and features such as the distance to the preceding aircraft and congestion levels. The experimental setup involved training XCSI on these data, followed by validation in a CA-based traffic simulator to assess the effectiveness of the generated speed control rules.

The results demonstrated that XCSI significantly improved operational performance, achieving a reduction of 10–20 seconds in average flight times and up to 5 fewer spacing adjustments for flights within 600 seconds before and after the arrival of rule-applied flights. Overall, the XCSI-based speed control led to a total flight-time reduction of approximately 1,801 s and fuel savings of 1,650.1 kg, resulting in an estimated annual economic benefit of $466,000. Notably, the algorithm extracted distinct speed control patterns across different airspace sectors and identified common rules applicable across multiple scenarios, demonstrating the flexibility and generalizability of the generated rules. Compared to traditional decision-tree methods, XCSI achieved a higher accuracy rate of 96.8% for speed control predictions, outperforming the 70.5% accuracy of decision trees.

Thus, this study highlights the potential of XCSI to support ATCos with practical and interpretable rules for managing en route air traffic, ultimately contributing to more efficient and robust arrival management. Future research should focus on extending this framework to encompass multi-step adaptive control, where aircraft actions can be continuously adjusted over time. Additionally, a detailed quantitative comparison with other machine learning methods, such as ensemble models or deep learning, could be conducted to evaluate their accuracy, efficiency, and interpretability in comparison to XCSI, enabling a more informed selection of the most suitable approach for real-world ATM applications. Additionally, validating the rules in more diverse airspace environments, including those with varying weather conditions and traffic patterns, would further strengthen the applicability and generalizability of this approach. To facilitate real-world implementation, conducting HITL simulations is essential for thoroughly assessing the system's effectiveness in realistic air traffic control environments, ensuring that the derived rules can be practically applied and further refined under dynamic conditions, while also considering human factors and operational constraints.

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